

LeaseOS / FieldRoute — Reconciliation & Forensic Audit

Date: 2026-09-16 · **Scope:** every ChatGPT-produced deliverable from Chats 1–5, the two upload archives, the Project Recovery lineage, and the two chats whose source never reached git (Chat 4, Chat 5).
Result: one master repository (424 refs), one reconciled integration branch that passes the clean-database CI gate end-to-end (== PASS == , exit 0, 3,035 tests), and 45 files recovered byte-faithfully from conversation transcripts.

1. What was delivered, in one table

Deliverable	Where	What it is
leaseos-master-2026-09-16.bundle	this package	Every ref from every bundle ChatGPT produced (423) plus integration/reconciled-2026-09-16 — the only branch that passes the gate
leaseos-reconciled-v22.21-src.zip	this package	Source tree at 515ce64 (868 tracked files; no node_modules)
leaseos-recovered-source-chat4-chat5-2026-09-16.zip	this package	The 45 transcript-recovered files in their <i>original</i> chat form, before renumbering — provenance for §5
LEASEOS_RECONCILIATION_AUDIT.md	this file	Findings, fixes, gaps
gate-final.log	this package	The full CI gate output for 515ce64

Lineage of the reconciled branch:

```
74d99de  main (pre-recovery, archived)
└─ 18444d4 → 5d3e42b (Project Recovery RC) → d9b3e51 → d6b8a5a →
068b29c → e9b2047 (ChatGPT tip)

└─ 2ccd305  gate: migrations + typecheck actually pass
```

```
4531847 recover Chat 4 + Chat 5 tranches (0118-0122)

8795756 tripwires re-baselined honestly

63e03a6 schema parity for 0122

56390c8 LEASEOS_CURRENT_STATE regenerated (v22.21)

fe3e137 four borrowed-name procedures given their own names

515ce64 field tests migrated to the 0110 signature protocol ←
integration/reconciled-2026-09-16
```

`main` was **not** fast-forwarded. It is eight commits behind the Project Recovery RC by the recovery plan's own design; promoting it is a decision for the repo owner, and the gate result in §3 is what that decision should rest on.

2. Archive integrity (what was verified, not assumed)

- `Chat_gpt_lease_os.zip` (431 MB, 29 nested zips) and the 8-part split `Chat_gpt_lease_os__1_.zip + .z01 - .z07` (~714 MB) both recombined and extracted cleanly.
 - After byte-identical de-duplication (≈55 -N-suffixed copies collapsed) **73 unique deliverables** remain; **every one passes its published SHA-256**.
 - One file named in a manifest is absent from both archives:
`LEASEOS_PROJECT_RECOVERY_BUNDLE_VERIFY_2026-09-14.txt`. Nothing depends on it.
 - All 10 git bundles pass `git bundle verify` and `git fsck`. Fusing them yields a single clean ancestry chain (all ancestry tests pass) — there are no divergent histories to reconcile at the git level.
 - `restricted/chat5-assessor-key` holds `LEASEOS_ASSESSOR_KEY_CONFIDENTIAL.html`. It is in the bundle. **Do not push that ref to a public remote.**
-

3. The gate — the finding that reframes everything else

ChatGPT's tip (`e9b2047`) was described in its own checkpoint as "513 TS/TSX source files parsed/transpiled with zero syntax diagnostics." That statement is true and meaningless: it was a *transpile*, not a typecheck, and **the CI gate had never been run on the recovery lineage at all**. Run `clean-database` on 2026-09-16 it failed at step 2.

Gate step	e9b2047 (ChatGPT tip)	515ce64 (reconciled)
0. Reserved slots 0016/0017	pass	pass
1. Clean database	pass	pass
2. Migrations	FAIL at 0108	pass (118 files)
3. Table parity	—	pass
4. Typecheck	21 errors	pass
5. Bare <code>protectedProcedure</code>	—	0
6. Test suite	— (never reached)	191/191 files, 3,035 passed, 0 failed, 3 skipped
7. Build / 7b portal / 7c machine gates	—	pass (36 external, 2 integration, counts pinned)
8. Current-state document	— (stale: 113/350)	regenerated (118 / 356 / 526)

3.1 Defects found in the ChatGPT tip, with root causes

#	Defect	Root cause
1	<code>0108_training_academy_hardening.sql</code> cannot apply	<code>scripts/apply-migrations.sh</code> carried the <i>verification</i> half of Chat 5's trigger fix (count the trigger) but not the <code>DELIMITER</code> half (make a compound <code>BEGIN...END</code> body reach the server intact). The gate could never pass step 2.
2	<code>contractorOperationsRouter.ts</code> : 14 procedures wired as <code>roleProcedure("contractor.write")</code> etc.	Procedures wired by permission name instead of registered procedure name — the exact defect Chat 3 documented as recurring. The file is also a single-line-per-procedure style the project rules forbid.
3	<code>integrationRouter.ts</code> : <code>ownershipAssign</code> , <code>ownershipList</code> , <code>loadSenseBindGateway</code> , <code>loadSenseCalibrate</code> wired under <code>integration.clientRegister</code> / <code>integration.inboundList</code>	Same defect, third occurrence — and <code>coreRecordOwnershipBoundary.test.ts</code> asserted the borrowed name , pinning the bug

#	Defect	Root cause
4	<code>for...of</code> over <code>Map / Set</code> in <code>loadSenseProtocol.ts</code> , <code>trainingAcademy.ts</code> ; top-level <code>await</code> in <code>worker.ts</code>	The project <code>tsconfig</code> has no <code>target</code> , so these need <code>downlevelIteration</code> / ES2022 that it does not set. Chats 1–3 never wrote these forms.
5	<code>integrationRouter.ts:229</code> passes a DB row where <code>LinearCalibration</code> is expected	<code>rSquared: number null</code> vs <code>rSquared?: number</code>
6	<code>readinessComposer.ts:251</code> <code>overrideAuthority: "safety"</code>	"safety" is not a role in the <code>DispatchBlocker</code> contract (<code>dispatcher manager administrator</code>). Chat 5's own bridge used <code>manager</code> .
7	<code>fieldDevice.test.ts</code> (2) and <code>fieldRuntime.test.ts</code> (3) fail	0110 changed the device protocol to P-256 SPKI keys + fresh <code>signedAt</code> + nonce + P1363 signature — and shipped with zero tests of the new path while leaving both existing suites on the fingerprint-only protocol
8	<code>reservedWordColumns</code> audit: a new column named with a MariaDB reserved word (<code>offset</code>)	0109 LoadSense migration, unreleased
9	<code>tenantIsolation</code> : untyped database handle in <code>loadSense</code>	post-recovery code
10	Four census tripwires stale (procedure counts, engine reachability)	additions without re-baselining

Every one of these is a defect the project's own gate would have caught in under ten minutes. **None was caught because the gate was never run.**

3.2 Design notes surfaced by the field-test migration (not changed; flagged)

- **0110 verifies the parsed input, not the wire input.** `receivePackage` verifies the signature over `input` after Zod applies `captureAuthorizationClaim: .default("unknown")`. The shipped client always sends the field (`syncEngine.ts:163`) so production is unaffected, but any client that omits a defaulted field signs a payload the server will never match. Verify over the bytes the device signed, or drop server-side defaults from signed schemas.

- **Unknown-key refusals are now thrown, not recorded.** Pre-0110 a package signed by a key the device never held landed as a *rejected row* ("refusals are rows", which the router's own doc comment still says). 0110 throws `UNAUTHORIZED` before writing anything. Defensible — an unattributable package cannot be recorded against a device — but it is a design change nobody wrote down until now.
- **Signature freshness is ± 10 minutes of server time.** A tablet whose clock has drifted more than that cannot push. Worth an explicit clock-skew UX in the field runtime.

4. Per-chat coverage — what reached git and what did not

Chat	Content	State in the reconciled branch
1	B17–B20: workflow engine, AI proposal/Secretary, dispatch concurrency, billing adjustments, worker endpoint, weigh-station pin	✔ present
2	v20.3→v22.19: measurement ladder, records vault, insurance/risk, exception centre, IFTA, fleet shop, capital assets, integration gateway, telematics, workforce, audit packages, spatial, money precision, rate resolution, road graph, communications, offline package	✔ present (<code>siteSignoff</code> lives as <code>siteCloseout.ts</code>)
3	0090–0106 AI stack (grounding, context admission, offline capability), document assistant, comms/route/document screens, calibration loop, retrieval probes, corpus hash	✔ present (<code>modelGateway.ts</code> is the renamed model router)
4	AI perimeter → 0093B knowledge/HOS tranche; B28 Phase 3A–3J widget-on-branch	✔ knowledge/HOS tranche recovered (§5) · ✗ Phase 3A–3J not recovered (§6)

Chat	Content	State in the reconciled branch
5	Academy 0087–0089: readiness bridge, sheet serials, TDG certificate contents, topic coverage, inspector requests, retention chain, printable tickets	✅ TDG/inspector/retention tranche recovered (§5) · ⚠️ academyReadinessBridge / sheetSerial still unmerged on fix/chat5-module- paths-and-vitest (§6)

Root cause of the Chat 4 loss (established last session, confirmed this one): Chats 3 and 4 both branched from the 0088 -era candidate and both numbered their migrations 0089–0093 . ChatGPT's recovery ingested Chat 3's full source and only Chat 4's markdown checkpoints; when it built the unified repo, Chat 3's numbers were already in place and Chat 4's source was never placed in the object store. It survived only in the conversation transcript.

5. What was recovered from transcripts, and how

Method: each Chat 4 / Chat 5 assistant turn that created or edited a source file was read from the conversation transcript; the file was reconstructed from the create_file text and every later in-turn safe-edit correction was applied in order (the transcript records each edit's exact OLD / NEW blocks). The staging copies in leaseos-recovered-source-chat4-chat5-2026-09-16.zip are the originals with chat-era migration numbers; the branch copies are byte-identical except where noted.

5.1 Chat 4 — knowledge and HOS (33 files, turns 69–94)

Chat 4 checkpoint	Files	Slot on branch
AI perimeter	server/_core/knowledge/admission.ts , perimeter.ts + 2 tests	—
Source licence gate	sourceGate.ts (511 Alberta record verbatim) + test; leaseos_511_alberta_license_gate.json filed beside it	—
0089 knowledge tables	0118_knowledge_source_registry.sql , schema block, repository.ts , DB test	0118
0090 provenance bridge	0119_hos_limit_provenance.sql , rulePromotion.ts + DB test	0119

Chat 4 checkpoint	Files	Slot on branch
0091 promotion ledger	0120_hos_rule_limit_history.sql , promotionLedger.ts + 2 DB tests	0120
0092 verification console	evaluationState.ts , citationGuard.ts , client/src/pages/HosVerificationConsole.tsx + route, verificationConsole.test.ts , Playwright spec (in staging zip)	—
Scope guard / P9	scopeGuard.ts + DB test	—
0093 cited path	hos.limitPromote in hosRouter.ts , registered in recordsAuthorization.ts , limitPromote.db.test.ts	—
0093A candidate correction	hosRuleSeeds.ts : south 780 / s. 12(1) cited; north 900 / s. 39(1) with HISTORY and STILL CONTESTED (on-duty) notes; hos.test.ts assertion inverted; federalCandidates.db.test.ts , federalCandidateCorrection.test.ts	—
0093B one door	hos.limitVerify closed (refuses, names limitPromote); promoteVerifiedLimit closed; oneDoorInvariant.db.test.ts	—

Invariants these restore, in ChatGPT's tip they were **absent**: no source is ingested without a stored licence assessment; chunking is gated on *reproduce*, not *search*; a regulatory figure reaches *verified* through exactly one door, which requires a named verifier, a binding instrument, a citation on a registered publisher's domain, a schedule scoped no wider than the reading, a plausible figure and three separate attestations; every promotion is an immutable ledger row; **P9 holds — no real HOS figure is verified anywhere in the tree.**

The only deliberate deviation from the transcript: `tdgTopicCoverage.ts` uses `Array.from(...)` instead of spread over a `Set` (tsconfig, see §3.1 #4).

5.2 Chat 5 — TDG evidence chain (12 files, turns 17, 47, 51, 59, 61)

Module	What it enforces	Slot on branch
<code>tdgCertificateContents.ts</code> + test (25)	s.6.2 (a)–(m) topic list verbatim from the 2026-06-21 consolidation; s.6.3(1) content gate; aspects derived from	pure module

Module	What it enforces	Slot on branch
	course coverage, never free text; 6.2(l)/(m) mode-conditional (the bug Chat 5's own tests found)	
<code>tdgTopicCoverage.ts</code> + test (24)	closed vocabulary; declaration must reconcile with the union of module coverage in both directions; approval binds to an order-independent fingerprint; seven distinct refusal codes	pure module
<code>inspectorRequest.ts</code> + test (20)	s.6.7 15-day clock from the later of dated/received; a partial package never reports complete; <code>irrecoverable</code> for material lost before the guards	pure module
retention chain guards (Chat 5 PENDING-5)	five BEFORE DELETE guards so a certificate cannot outlive its course version, modules, content blocks, assessment or statement of experience	0121
inspector requests table (Chat 5 PENDING-6)	<code>academyInspectorRequests</code> with <code>dueAt > requestDatedAt</code> enforced	0122

These three modules are **declared unwired** in `engineReachability.test.ts` with reasons: the issuance and inspector-request procedures that consume them are the next Academy checkpoint. Chat 5's PENDING-2/3 (assessment sheet registry + integrity triggers) and PENDING-4 (`tdgTopicCoverageHash` / `tdgTopicReviewedHash` columns) are in the staging zip and **not** on the branch — they depend on the still-unmerged `sheetSerial` work.

6. Remaining gaps (in priority order)

- B28 Phase 3A–3J** (Chat 4 turns ≈40–68): the widget engine's integration onto the real branch — `boardSemantics`, `widgetService`, `widgetRegistry`, the `/widgets` route, the three widget integration tests, accessibility spec, licence-gate script, template adoption. The branchless engine source is preserved on `candidate/b28h-widget-engine` (`bc91356`); only the integration layer must be mined from the transcript the same way §5 was. Next free slots: **0123+**.
- Chat 5** `academyReadinessBridge` / `sheetSerial` / `sheetSerialAllocator` on `fix/chat5-module-paths-and-vitest` (`54965ee`): import path fix, numbered-copy filenames, and the PENDING-2/3/4 SQL to port. Then wire the three §5.2 modules into `trainingAcademyRouter` and remove them from `DECLARED_UNWIRED`.

3. **limitPromote** **separation of duties**: `profileVerify` requires a second person; `limitPromote` does not (needs `recordedByUserId` on `hosRuleLimits` — its own migration). Named by Chat 4's 0093B; still open.
4. **CA_FEDERAL_NORTH60.daily_on_duty_minutes** still carries the southern 840 — marked STILL CONTESTED in the seed. A verifier with s. 39 open decides it.
5. **Native field runtime** (Capacitor/SQLite/keystore/camera/GPS/biometric), **provincial routing runtime**, **multi-company tenant isolation**, **LoadSense hardware ingestion**, **LICENSE file** — unchanged from the 2026-09-14 gap matrix; none was in scope here.
6. The three §3.2 design notes.

7. How to use the package

```
git clone leaseos-master-2026-09-16.bundle leaseos && cd leaseos
git checkout integration/reconciled-2026-09-16
pnpm install --frozen-lockfile
export DATABASE_URL="mysql://leaseos:leaseos@127.0.0.1:3306/leaseos_ci"
bash scripts/ci-gate.sh          # clean database, 8 steps, exit 0
expected
```

Verified on: MariaDB 10.11.14, Node 22, pnpm 10.4.1. The gate drops and recreates the database it points at.

8. What this audit does not claim

- It does not claim the recovered code is *correct* beyond what its own tests assert; it claims the code is what Chat 4 and Chat 5 actually wrote, plus their own corrections, plus the three fixes in §3 needed to compile under this tree's `tsconfig`.
- It does not claim any regulatory figure is verified. Every HOS value in the tree is an unverified candidate or a fixture, and the tests say so.
- It does not claim ChatGPT's post-recovery features (LoadSense gateway, contractor payables, security runtime) are *complete*; it claims they now typecheck, migrate and pass their tests, and that four of their procedures were wired under other procedures' names until today.

9. Package checksums (SHA-256)

```
b1577ddfcfca1758099936c370d9cc1627dd141a3d7cad395d00423b0a9f6a23  
leaseos-master-2026-09-16.bundle  
3347dd71e86c77e7ae41d4b1c0beafd4ed6957dfe9afd363628adee7b9bed05f  
leaseos-reconciled-v22.21-src.zip  
316903d64cd86cecb2c15094f1fbfc494038b1033894319d8f3a2f10ad2223fa  
leaseos-recovered-source-chat4-chat5-2026-09-16.zip  
10ef97b96ad531906f0463324cf5dca925660ebdcb3865d36e7f432f990a9445 gate-  
final.log
```

Bundle: 424 refs, `git bundle verify` clean, round-trip clone of
`integration/reconciled-2026-09-16` yields 868 tracked files at `515ce64`.